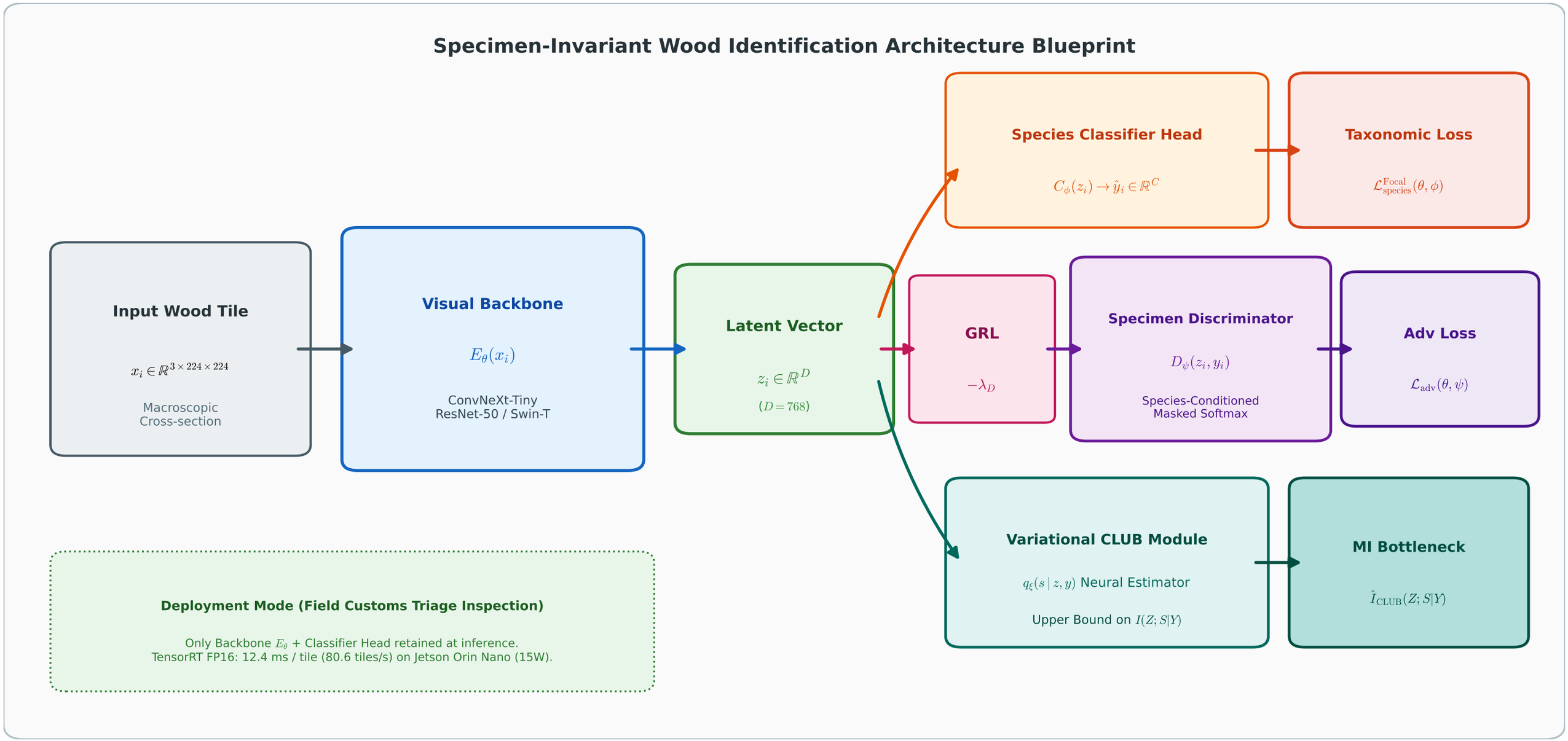

Specimen-Invariant Wood Identification Architecture Blueprint
Input Wood Tile
$x_i \in \mathbb{R}^{3\times 224\times 224}$
Macroscopic
Cross-section
Visual Backbone
$E_\theta(x_i)$
ConvNeXt-Tiny
ResNet-50 / Swin-T
Latent Vector
$z_i \in \mathbb{R}^D$
$(D = 768)$
Species Classifier Head
$C_\phi(z_i) \to \hat{y}_i \in \mathbb{R}^C$
Taxonomic Loss
$\mathcal{L}^{\text{Focal}}_{\text{species}}(\theta, \phi)$
GRL
$-\lambda_D$
Specimen Discriminator
$D_\psi(z_i, y_i)$
Species-Conditioned
Masked Softmax
Adv Loss
$\mathcal{L}_{\text{adv}}(\theta, \psi)$
Variational CLUB Module
$q_\xi(s \mid z, y)$ Neural Estimator
Upper Bound on $I(Z; S|Y)$
MI Bottleneck
$\hat{I}_{\text{CLUB}}(Z; S|Y)$
Deployment Mode (Field Customs Triage Inspection)
Only Backbone E_θ + Classifier Head retained at inference.
TensorRT FP16: 12.4 ms / tile (80.6 tiles/s) on Jetson Orin Nano (15W).